Topics in Geobiology

ESS/EPS 208
Spring Quarter 2026
Braun Geology Corner 320
Wednesdays 3:00pm - 4:30pm

Primary Instructors

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Course Description

The interdisciplinary field of geobiology strives to understand the interactions and coevolution between life and Earth. This broad field explores questions in prebiotic chemistry, biogeochemical cycles, major mass extinctions, modern and ancient microbiology, metabolic innovations and extreme living, planetary habitability and biosignatures, climate through time, and more. If you don’t know what some of these topics are or the volume of content feels overwhelming – that’s okay! This course is designed to demonstrate how scientists expand our mindset through discussion and exploration of new topics in a structured format.

By taking this class you agree to be respectful and accommodating to different view-points and perspectives. The community of this class will be welcoming of all cultural backgrounds, diversity in ability and thought, and self-expression.

Learning Goals

1. **Survey** the core pillars of Geobiology (outlined above) through peer discussions and critical reading
2. **Gain** familiarity and confidence in engaging with literature outside domain specific knowledge
3. **Interrogate** the conclusions of primary literature and consider alternative interpretations based on available evidence
4. **Evaluate** the clarity of data presentation, methods of collection, interpretation, and the fair use of scientific data.
5. **Strengthen** scientific communication and interpersonal skills through the facilitation of group discussions.

You will be successful in this course if you discover a new interest, make connections with someone outside of your regular sphere of influence, expand your worldview, and/or feel more confident talking openly about science. Most importantly, learn something in a low-stakes environment.

Course Format and Schedule

Format

The course is structured as a weekly discussion seminar. Each enrollee will sign-up for a discussion leader slot within first week of quarter on a calendar. At least one week before their presentation date, discussion leaders will choose a paper to explore in their session and upload it to Canvas to allow adequate access to peers. All participants are expected to read and digest the paper before coming to the seminar. We will offer resources for critical reading skills on Canvas.

Discussion leaders will prepare discussion in the format of their choosing (e.g., slideshow, scrolling through paper, whiteboard etc.) in which they provide background context for the paper, their critical thoughts and figure slides to be discussed by the group. A list of questions to help motivate discussion will also be provided on Canvas. Instructors will provide resources for how to lead a scientific discussion on Canvas and act as facilitators for each discussion.

The first day of instruction will be introductions, an (optional) ice-breaker, followed by a general overview of Geobiology and principles of scientific literacy by the co-instructors. The remainder of the course meetings will be dictated by your enthusiasm for science.

Course Content

Paper selection and leading discussions

Papers should be peer-reviewed, primary research articles that are of interest to you. Papers can cover any topic in geobiology. A list of suggested papers will be available on Canvas (and below) but you are free to present a paper not on this list. As a discussion leader, you will lead the conversation about the paper and each of the figures

*If you require accommodation for your presentation, please reach out to Ashley at least 1 hour before class to coordinate.

Suggested Reading

Geobiology is a broad discipline. Some participants will already be familiar with foundational literature and will have interests in frontier science; some will be totally new and desire to gain familiarity with the foundations of a particular sub-field. Either way, introduction to the seminal papers in the different disciplines represented are a good place to start/expand your journey into geobiology.

The papers below, organized by their primary focus, fundamentally changed the perspective of their respective disciplines, and you are free to present them as your discussion. Notice, a core element of most papers is typically the cross-disciplinary approach that bridged connections between fields. A possible discussion topic: *How do some of these original papers hold up to modern-day scrutiny?*

Microbiology

Falkowski et al (2008)	The microbial engines that drive Earth's biogeochemical cycles	https://doi.org/10.1126/science.1153213
Sagan 1967	"On the Origin of Mitosing Cells"	https://pubmed.ncbi.nlm.nih.gov/11541392/

Geochemistry

Lyons et al (2014)	The rise of oxygen in Earth's early ocean and atmosphere	https://doi.org/10.1038/nature13068
Alvarez et al (1980)	Extraterrestrial Cause for the Cretaceous-Tertiary Extinction	DOI: 10.1126/science.208.4448.109
Canfield (1998)	A new model for Proterozoic ocean chemistry	https://www.nature.com/articles/24839

Nutrient Cycling

Redfield (1934)	The biological control of chemical factors in the environment	https://ebme.marine.rutgers.edu/HistoryEarthSystems/HistEarthSystems_Fall2008/Week4b/Redfield_AmSci_1958.pdf

Evolution/Paleobiology

Gould and Eldride (1972)	Punctuated equilibria: the tempo and mode of evolution reconsidered	https://www.uv.mx/personal/tcarmona/files/2010/08/Gould-y-Eldredge-1977.pdf
Raup and Sepkoski (1982)	Mass Extinctions in the Marine Fossil Record	https://www.science.org/doi/pdf/10.1126/science.215.4539.1501

Planetary Habitability

Miller (1952)	A Production of Amino Acids Under Possible Primitive Earth Conditions	https://www.science.org/doi/10.1126/science.117.3046.528
Kasting et al (1993)	Habitable Zones around Main Sequence Stars	https://doi.org/10.1006/icar.1993.1010

Earth System Modeling and Climate

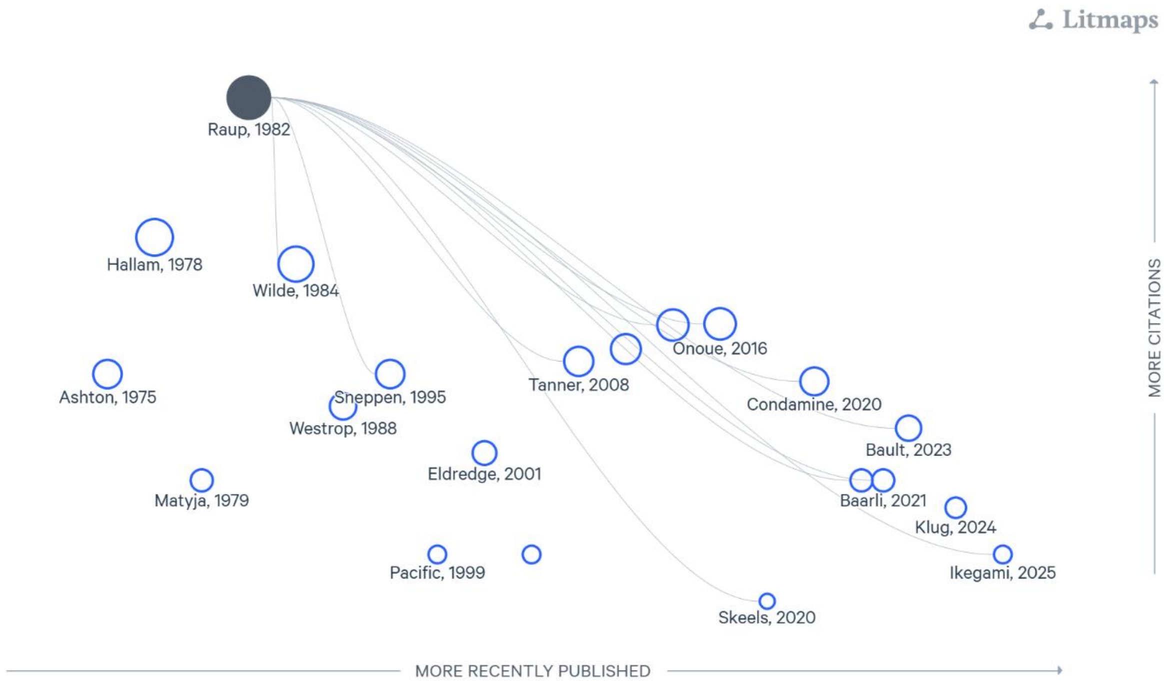
Cloud (1972)	A working model of the primitive Earth	https://doi.org/10.2475/ajs.272.6.537
Kirchner (2002)	The Gaia Hypothesis: Fact, Theory, and Wishful Thinking	doi.org/10.1023/A:1014237331082
Kirschivink (1992)	Late Proterozoic Low-Latitude Global Glaciation: the Snowball Earth	https://web.gps.caltech.edu/~jkirschvink/pdfs/firstsnowball.pdf

Metabolism and Physiology

Kleiber 1932	Body size and metabolism	https://ecol586.wordpress.com/wp-content/uploads/2018/01/kleiber1932-2.pdf

Finding Reading Topics

To expand your scope, there are several open-source tools to help you navigate the vastness of scientific literature. Some recommended sources are network or cloud-based visualization tools (e.g., Litmaps, Inciteful, Connected Papers) for paper discovery. We will talk more about this in the first session.



Course grading

1 unit
Satisfactory / No Grade

For enrollees, receiving a Satisfactory for this course will depend on attendance, leading one group discussion during the quarter, and active participation in discussion each week. We will talk about what active participation and those expectations mean the first day of class. Auditors will not receive a grade but are encouraged to engage in discussions.

Policies

Contact and Email Policy

Each instructor will have a personal preference for outside-class lines of communication. Ashley aims to respond to email requests within 24 hours and is welcome to office-walk-ins / walk-and-talks / coffee break between 9 and 11 am weekdays room 320.

Attendance policy

This course requires in-person attendance. However, we understand that there are many reasons, including illness or caretaking, that make attendance difficult. Hybrid access to the seminar can be provided on an individual basis. Please contact the primary instructors by email at least an hour before the seminar to work out online access. If you are unwell, please do not attend class and alternative arrangements can be made.

AI Use Statement

Considering the diversity of educational backgrounds and disciplines represented in this course, we acknowledge that AI can serve as a valuable tool. It has the potential to “raise the floor” by helping all students reach a shared baseline of knowledge, and to “raise the ceiling” by expanding the breadth of topics students can explore. While we do not discourage the use of AI for these purposes, **developing scientific literacy and building familiarity with the primary literature are fundamental objectives of this course,** which cannot be achieved through AI.

Accessibility

It is our goal to create a safe space for the geobiology community to explore new ideas and learn from each other, regardless of identity or ability. If there are any accommodations or adjustments that would help you thrive in this course, please reach out to the primary instructors and we will work together to find solutions. The Office of Accessible Education is another resource available:

Stanford is committed to providing equal educational opportunities for disabled students. Disabled students are a valued and essential part of the Stanford community. We welcome you to our class.

If you experience disability, please register with the Office of Accessible Education (OAE). Professional staff will evaluate your needs, support appropriate and reasonable accommodations, and prepare an Academic Accommodation Letter for faculty. To get started, or to re-initiate services, please visit oae.stanford.edu.

If you already have an Academic Accommodation Letter, we invite you to share your letter with us. Academic Accommodation Letters should be shared at the earliest possible opportunity so we may partner with you and OAE to identify any barriers to access and inclusion that might be encountered in your experience of this course.